preamble, Methods 3, 2026

2026-09-08

REMEMBER: This preamble is **NOT** part of your portfolio, but is a prerequisite for doing the portfolio

Preamble - ***GitHub***,***Python***, ***Conda***

The goals of the preamble are:

1) create a *Conda* environment that contains the *Python* packages that we need. Note that we are not creating an R environment - I expect you to maintain your own
2) install your R-packages
3) get a *GitHub* profile to the *GitHub* classroom such that you can get the materials of the course and hand in assignments

1) What is ***Conda***?

Conda is a package management and environment system.

What is a package?

(from ChatGPT)

- *Python*: Packages are collections of Python modules. They are often distributed as .whl or .tar.gz files. Examples include *numpy*, *pandas*, and *requests*.

- *R*: Packages are collections of R functions, data, and compiled code. They are distributed as .tar.gz files. Examples include *ggplot2*, *dplyr*, and *shiny*.

Why environments?

We use environments to create **isolated**, **shareable** and **conflict-free spaces** for our projects, e.g. the *methods3_2026* that we are going to create now won't interfere with other projects that we may be running

Installing ***Conda***: miniconda vs. anaconda

Section can be skipped if you already have miniconda installed

I recommend installing *Conda* using the **miniconda** distribution. The **anaconda** distribution comes with many pre-installed packages, which may lead to conflicting packages when creating environments.

Command line install (preferred method)

Link: https://docs.anaconda.com/miniconda/#quick-command-line-install

(If you already have **anaconda** installed, you may prefer keeping it to not create any conflicts)

On Mac and Linux, use the terminal; on Windows use the Anaconda Power Prompt that comes with installing anaconda

Create *method3_2026* environment

```
cd <path_that_contains methods_environment.yml> # go to folder with methods3_environment.yml file
conda env create -f methods3_environment.yml # create environment
conda activate methods3_2026 # activating your newly created environment
```

First goal achieved

You are now in an environment that is **isolated** from all *Python* that you may have installed at earlier date.

The **shareable** *methods3_environment.yml* file contains the recipe for the environment and *Conda* makes sure that the installation is **conflict-free** in terms of dependencies.

Installed packages:

```
cat methods3_environment.yml
```

```
name: methods3_2026
channels:
- defaults
- conda-forge
dependencies:
## Python version
- python>=3.13.7

## Python packages
- pip=25.2
- scikit-learn=1.7.1
- matplotlib=3.10.5
- numpy=2.3.1
- scipy=1.16.1
- pandas=2.3.2
- seaborn=0.13.2
```

2) R packages to install

We are going to be dependent on the packages `lme4` and `fields`.
Please install as you usually do

3) Get your github working

(If you don't have a *GitHub* account sign up at www.github.com)

Go to https://github.com/Methods-3/Methods-3-F26/tree/main/assignments/preamble and download *preamble_test.Rmd* and do the assignments in there

This concludes the preamble